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Bangla Sign Language Gesture Recognition System Using CNN Model

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***Abstract* - Sign language is a form of communication used by the deaf and dumb. However, integrating them into society is challenging because most people do not speak their language. It is a method of generating and disseminating ideas, information, points of view, facts, and sentiments to establish a shared understanding. Unfortunately, almost 5% of the world's population is not blessed with the ability of verbal communication. The situation is the same for Bangladesh. This paper presents a simple, low-cost Bangla sign language (BdSL) recognition model with maximum accuracy to convert signs into Bangla text. We've manually captured images for Bangla sign language following the BdSL model and implemented the dataset to train the system and show output. We used neural networks as a deep learning method to train individual signs to achieve our goals in the suggested model. Using a CNN, we separated the dataset into two parts: training data and test data. The action is mapped to the relevant text in the training data using image processing techniques, and neural networks, so raw images or videos are turned into text that can be read and comprehended with accuracy near 92%. Finally, the system will turn the user's signs and motions into text. It is referred to as "sign language recognition." In the future, This research can also open doors to numerous other applications, like sign language tutorials or dictionaries, and can also help the deaf and dumb search the web or send emails more conveniently. We hope this system will be an excellent opportunity for people who have speaking or hearing disabilities by helping them explore more.**

Keywords - BdSL; CNN; structure-building; training data; Deep Learning; Recognition

I. INTRODUCTION

Sign language is the only way for deaf and hard-of-hearing persons to communicate and express themselves [1]. Hard hearing is a loss of aptitude that impairs their ability to tune in and castrates their hearing, and dumb refers to a lack of skill that impairs their ability to communicate and renders them speechless [2]. 2.4 million People utilize sign language in Bangladesh, yet most do not understand it [3]. Bangla is the mother tongue of both the general public and the disabled. Hence there should be a universal medium for everyone. Disabled persons fall behind the general public due to communication issues [4]. This problem can be solved by using a versatile medium.

On the other hand, the United States has been using its own created sign language pattern known as American Sign Language (ASL) [5]. We have to reduce the communication gap so that hard of hearing people can express themselves more easily. Only a few people have worked with the Bangla sign language concept. Deaf and mute persons find it challenging to learn their signs because no relevant model can be used as a communication strategy. We are motivated to propose the system as it is critical to develop a system that converts sign language to text, allowing deaf people to communicate with the general public.

II. LITERATURE REVIEW

Language developments and an emerging body of research in education have validated the NAD's historical views on ASL and its value in educating deaf and hard-of-hearing children. Children's natural ability to learn ASL should be recognized and used to enhance their cognitive, academic, social, and emotional development. As a result, the NAD advocates for a bilingual approach for deaf and hard-of-hearing children. Deaf and hard-of-hearing children must have the right to early and comprehensive exposure to ASL as a primary language alongside English.

According to the National Institute on Deafness and Other Communication Disorders (NIDCD), the 200-year-old American Sign Language is a complete, complex language (letter gestures are only a tiny part). Still, it is the primary language of many deaf North Americans. As a result, developing a system that recognizes sign language will enable the deaf and hard-of-hearing to communicate more effectively using modern technologies. This article will look at various CNN architectures and how they perform when classifying Sign Language [6].

People with hearing and vision loss are excluded from participation in their family environment and society, their rights are ignored, and their needs are not met. This rejection can lead to a subhuman existence if it is not recognized and understood. It prompted CDD to intervene to raise awareness and understanding of deafblindness and provide support to people and families affected to uphold fundamental rights and give a voice to the most marginalized disabled people[7]. Previous research in referential communication has found that typically developing children progress in communicating a specific referent to a listener. In one paradigm, subjects were given lists of word pairs, one of which was designated as the referent. It was demonstrated that communicating about

referents found in word pairs that were somehow associated was more difficult than speaking about referents found in different word pairs. The current study extended this methodology to children with learning disabilities. There were few differences in the groups' performance on these referential communication tasks. The importance of vocabulary and concept development in referential communication is one of the implications [8]. A further analysis limited to the deaf group revealed that these results were not affected by the age of subjects' acquisition of ASL. This research adds to our understanding of the processes by which the handshape component of sign forms is recognized in a sign language, the role of language experience, and the extent to which these processes can be explicitly considered linguistic [9]. This paper investigates the relationship between movement, motion, and thought in humans and nonhuman primates and its role in language evolution. Human gestures (mainly symbolic gestures) may provide a specific link between action and mental representation. It is kinesthetically close to the action while also being symbolic. Nonhuman primates communicate frequently and in a variety of ways.
On the other hand, their gestures resemble incomplete movements and lack the symbolic elements that characterize many human gestures. We contend that gesture played an essential role in human evolution's transition to symbolic thought and language, following a cognitive leap that allowed the motion to incorporate representational elements [10]. Hearing loss is defined as diminished hearing caused by a variety of factors. It can be inherited or acquired later in life. Mild hearing loss to profound hearing loss is all possibilities [11].

The following paper presents an approach for recognizing static fingerspelling in Japanese Sign Language on RGB images. Two 3D articulated hand models were created to generate synthetic fingerspellings and supplement a simple hand gestures dataset. The first method used advanced graphics techniques to rasterize photorealistic gestures using a skinned hand model. A modified Generative Adversarial Network was used to post-process gestures rendered using more straightforward lighting techniques in the second approach. A hand segmentation term has been added to the GAN's loss function to avoid the generation of unrealistic fingerspellings. The proposed ResNet34-based segmentation network was used to segment the hand in images with complex backgrounds. An ensemble of fine-tuned and trained-from-scratch neural networks recognized the fingerspelled signs. The experimental results show that sufficient training data can achieve a high recognition rate on RGB images. The JSL dataset with pixel-level hand segmentations can be downloaded [13].

Hearing and speaking impairments are extremely debilitating handicaps for humans. People with such impairments frequently communicate using sign language, which is very different from spoken and written language, often leading to communication difficulties when a translation medium is not readily available. This causes a significant problem in the affected people's daily lives, particularly in Bangladesh. A simple, accurate, and portable translation medium that allows two-way communication between an average person and an impaired person is required to address this issue. Speech recognition and conversion to sign language is a critical component of such a medium, especially since Bangla is a complex language to work with. This paper describes the development of a smartphone application that provides a two-way communication platform using a speech-to-sign language system and a text-to-speech system. The application's testing yielded satisfactory results, with an average accuracy of 84.71 percent in various test environments [14].

This paper proposes an Artificial Neural Network-based hand sign language recognition framework for various Bangla alphabets (ANN). The boundary edge of the hand sign area is extracted using the canny edge detector, and the hand sign region of interest is extracted (ROI). Finally, features are removed from the hand sign ROI (FCC) using Freeman Chain Code. The ANN is used for training and categorizing the images of hand signs. The proposed method is tested with various hand sign images, and the results are presented to show its efficiency and effectiveness [15]. A real-time computer vision-based Bengali Sign Language (BdSL) recognition system is presented in this paper. The system recognizes the likely hand in the captured image. The system employs Haar-like feature-based cascaded classifiers to detect the writing in each frame. The system extracts the hand sign from the detected hand area based on the Hue and Saturation value corresponding to human skin color. The system converts the hand sign to the binary image after normalization. The binary images are then classified using the K-Nearest Neighbors (KNN) Classifier by comparing them to pre-trained binary images of hand signs. The system can recognize six Bengali vowels and thirty Bengali consonants. The system is trained using 3600 (3610) training images. Each of the ten signers performed ten signs for each corresponding Bengali alphabet. It is tested using another 3600 (3610) pictures of ten signers. The system achieved 98.17 percent vowel recognition accuracy and 94.75 percent recognition accuracy for consonants [16].

Researchers are having a difficult time collecting hand gesture data for sign language. This article examines Ishara-Lipi, the first complete isolated characters dataset of Bangla Sign Language (BdSL). It will increase interaction between the hearing-impaired community and the general public. The dataset contains 50 sets of 36 Bangla primary sign characters collected with the assistance of various deaf and available volunteers. Bangla Sign Language sign characters have six vowels and thirty consonants, allowing them to fingerspell all Bangla words. After removing errors and pre-processing, 1800 character images of Bangla Sign Language were included in the final state of the Ishara-Lipi dataset. This dataset could be used to create a computer vision-based system or any other type of system that allows users to look up the meaning of a BdSL sign [17].

We will now create a more assertive approach to image classification, which will naturally extend to entire Neural Networks and Convolutional Neural Networks. The method will consist of two major components: a score function that converts raw data into class scores and a loss function that quantifies the agreement between predicted scores and ground truth labels. The problem will then be recast as an optimization problem in which we will minimize the loss function concerning the score function's parameters [18].

III. Methodology

Because the computer system cannot recognize the sign, any type of sign language identification is extremely difficult. The researchers put in a lot of effort and used various methods to identify the signal using a computer system.

Our proposed system obtained an open-source dataset from the BdSL recognition model that is open to the public. 'Ishara-Bachon' and 'Ishara-Lipi' are the digits and character dataset names, respectively [19]. We initially looked at some previous work related to our research, such as segmentation-robust modeling for sign language recognition [20] and sign language and human activity recognition [21]. Still, we primarily used our technique for Bangla sign language recognition. Based on the Microsoft Research group's publication on visual document analysis [22], we studied the application of convolutional neural networks for feature learning. However, we believed that a more uncomplicated strategy would still produce competitive accuracy results.

A. Block Diagram

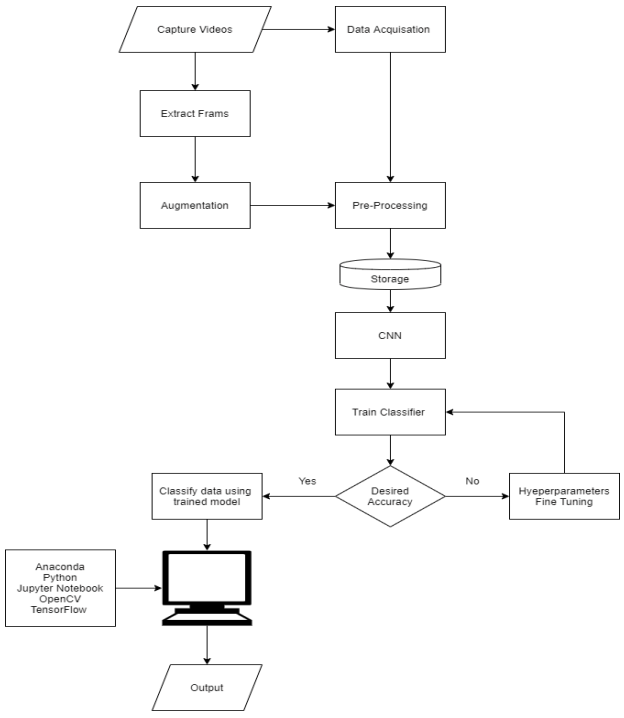


Fig. 1. Data Flow Diagram

The overview of the proposed system is illustrated below with data acquisition and pre-processing. The video frames will be submitted as the input to the system, and the output will be the predicted sign digit presented in the Bangla language. Video frames as pictures will be augmented to change the dataset, and the model was trained using CNN. The model will be stored and loaded with OpenCV to recognize the signs in real-time.

B. Use Case Diagram

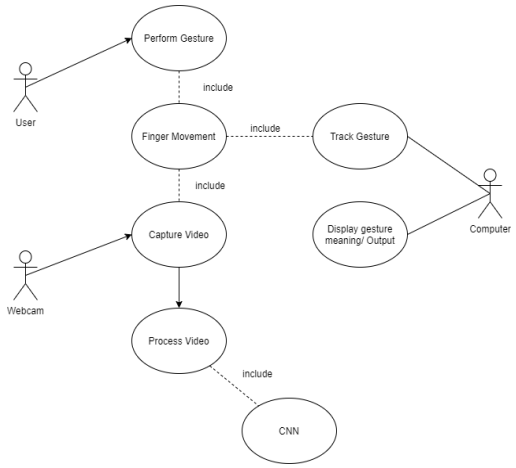


Fig. 2. Case Diagram

Use case diagrams are also known as behavior diagrams. They depict behaviors that specific systems (subjects) should or may do in partnership with the platform's multiple stakeholders (actors). To the actors or other users of the system, each user should deliver some measurable and valued output. So the primary workflow of our system will be the followings:

1) Capture Video: The primary purpose of recording a video will be to process it. The webcam is the main character, while the user is the supporting character. The user should have or have access to a webcam. The live stream will be collected here, and operations will be carried out. If the program fails, there'll be an exception.

2) Process Video: The purpose here is to process the videos that have been received. Webcam is the main character. A webcam is used to capture live video, transformed into photographs. Various operations are carried out to improve related properties and gesture actions.

3) Display Gesture Meaning: The output/meaning based on tracking and gesture recognition will be shown here. The computer is the main character, with the user and camera as minor characters. The user must have or have access to a camera as a precondition. The network will be used to recognize the improved photos.

C. The optimization algorithm

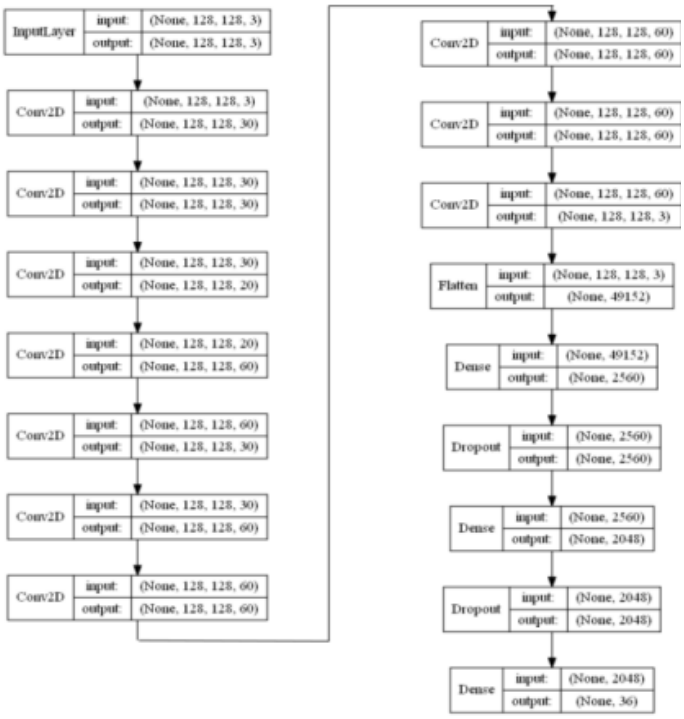


Fig. 3. The neural network architecture is depicted in this diagram.

The optimization algorithm chosen can significantly impact the outcome in Deep Learning. Computer vision and learning go hand in hand. According to a recent study, the cross-entropy function outperforms classification and mean square errors. The weight changes don't get smaller and smaller due to cross-entropy error, so training isn't likely to stall. Using an automated Learning Rate reduction strategy, make the optimizer converge faster and closer to the global minimum of the loss function. The learning rate is the number of steps taken to walk through the least amount of loss. If a faster learning rate is used, the system will soon converge and become locked in a local minimum rather than a global minimum. After each epoch, the model dynamically decreases the learning rate by monitoring validation accuracy to maintain the benefit of quick computation time with a high Learning Rate.

IV. Implementation

A. Dataset Description

Had the data for training and testing came from a dataset we created by ourselves. A digital camera captured 1005 samples of the 36 signed Bangla letters and the accompanying depth data. A person signs the corresponding letter directly facing the Digital camera in each piece. The following is a representation of some of the dataset's unprocessed raw images:

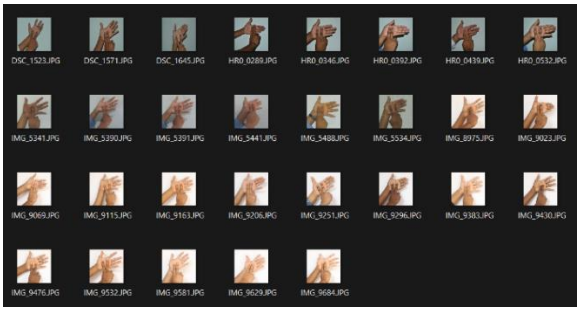


Fig. 4. Example Images in Dataset

B. Technical Approach

The PILLOW library, an image processing toolkit, and the sklearn, the decomposition package, valuable for its matrix optimization and decomposition functionality, were used to pre-process the data. The fundamental objective has been broken down into four stages- Image Capturing, Data Labeling, Image Cropping, Resizing, and Converting to Gray-scale. We started by resizing the image to 64*64 pixels, then converting RGB to grayscale.

a) Capturing Data: Open-source data is being gathered, and photographs of bare hands are being taken. The white background is chosen as the background color, and we then take one shot at a time. Because of the high resolution, a DSLR camera is used to photograph hand signs. We aim to reduce white space from the hand's edge while collecting photographs.

b) Labelling Data: Labeling data is critical for reducing noise in capturing and improving image quality. We'll write our script after gathering data. Some BdSL signs, for example, are very similar to one another. Images of this type of sign look familiar and add to the sense of experimentation. We'll do it by categorizing the signs individually. Different classes/folders exist for different signs. For digits, ten different directories (numbered 0, 1, 2, 3, 4, 5, 6, 7, 8, 9) were created. On the other hand, Bangla has roughly 49 characters, but not in sign language. There are only 36 characters in Bangla Sign Language. As a result, we have 36 characters after naming them with a numeric convention ranging from 1 to 36. (1, 2, 3...9). After that, we organize the photographs by categorizing them into distinct classes. The naming scheme for all characters is shown below.

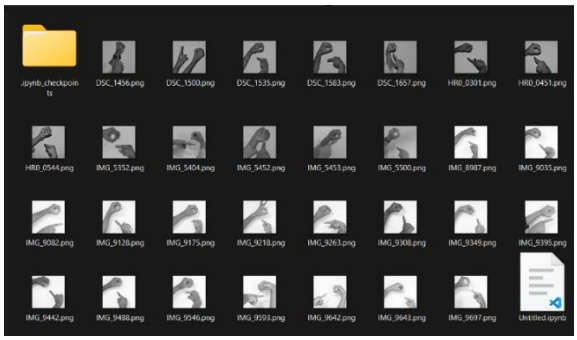


Fig. 5. Image for Segmentation

c) Cropping Data: The captured image cannot be used without cropping for sign recognition purposes. Cropping is required to obtain an idea that may be used for further experimentation. We're manually cutting photos to remove noise and keeping track of the height and width ratios for further processing. The hand region is cropped from the images. Cropping is used to reduce as many white areas as feasible.

d) Resizing Data: All photos will be scaled in a standard format using a python cv2 script to make our data more useful. We're writing a script that will go into each folder containing images and do the identical operation on each image. The image will be resized to 64 * 64 pixels by the scripts.

e) Converting Data to Grey-Scale: The script will convert RGB to grayscale after scaling. All photos will be prepared in a uniform format, with the same pixels in height and width. All sign photos will be reduced to 64 * 64 pixels and resized. There will be 36 characters, with six Bangla vowels and thirty consonants. All images will be saved in the.jpg data type. The photos will be collected, resized, and transformed from RGB to grayscale.

C. Application and Performance

We have tried to implement some popular methods for quick image segmentation, particularly hand segmentation. Using a Canny edge detector to discover relevant "things" in the camera's field of vision was one of the ways we investigated. To make a solid, continuous mask, the edges were dilated, and then all remaining holes in the mask were filled.

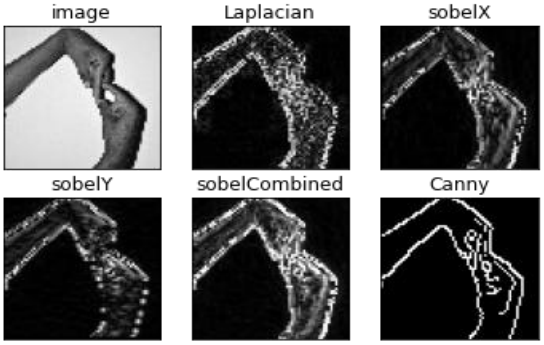


Fig. 6. cannyEdgeDetector Implementation

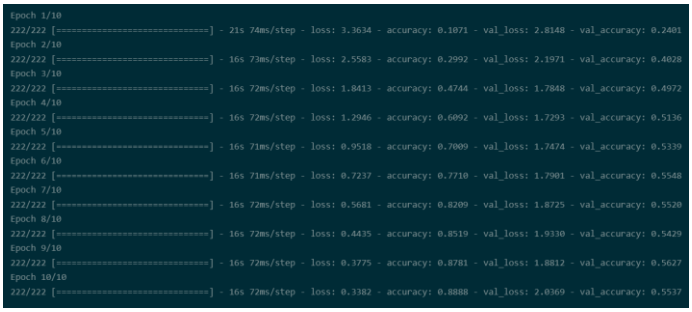


Fig. 7. Value Accuracy from training

After that, just the significant portions were photographed to eliminate all background clutter objects. This method is based on the notion that the most oversized items in segmentation are also the ones of most significant interest. We used solely color information to test the two algorithms for skin segmentation. The initial method involves fitting a 2D Gaussian curve to the skin color and then estimating the likelihood of a given color pixel being skin using this fitted Gaussian. First, we gathered skin patches from 40 different photographs found on the internet. Each skin patch was a rectangular skin region that was contiguous. Skin patches were obtained from people of various races for our model to accurately forecast skin regions for a wide range of skin colors. After that, the colors were adjusted as follows:

$$r = \frac{R}{R+G+B}, b = \frac{B}{R+G+B}$$

It is ignored because the g component is linearly dependent on the other two. The 2D Gaussian mean and covariance matrix (with r and b as the axes) is calculated as follows:

$$\text{Mean } m = E[x], \text{where } x = [r, b]^T$$
$$\text{Covariance } C = E[(x-m)(x-m)^T]$$

The likelihood of skin for any pixel in a given test image can be calculated using this Gaussian fitted skin color model. If the chromatic pair value of a pixel is (r, b), the likelihood of skin for that pixel is:

$$Likelihood = e^{[-0.5(x-m)^T C^{-1}(x-m)]}, where\ x = [r, b]^T$$

Finally, we used a threshold to determine if it was skin or non-skin. However, this method did not produce consistently good results and failed to detect weakly light skin areas. These alarming results didn't fulfill the hand segmentation quality criteria we needed to acquire consistent feature extraction from the model's learning layer

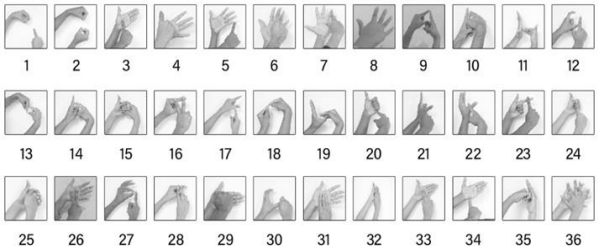


Fig. 8. Visualization of processed Database Images

We filtered out the hand region and deleted the face/other background pixels that would have been discovered using the acquired depth data after detecting the skin regions with great accuracy. The hand was presumed to be the focus of interest in each given frame and thus the closest skin-colored item in the camera's view. Each image was cropped and reduced to fit into a 64x64 pixel square. We then used this segmentation model to build a supplementary dataset containing hand gestures for each BDSL alphabet's 36 letters.

D. Classification and Feature Learning

To accomplish the actual recognition training phase of the project, the extracted data from hand pictures were loaded into an autoencoder. An unsupervised learning algorithm is implemented at this stage. All of the data samples are fed into the sparse autoencoder. The segmentation block's input data consists of 64x64 pixel pictures. Initially, a sparse autoencoder with a 64x64 node input layer and a 100 node hidden layer is chosen. The cost function was optimized using L-BFGS. This was running for around 400 iterations to get estimates of the weights. The autoencoder now knows a collection of features that are similar to edges.

The Softmax classifier uses the cross-entropy loss. The Softmax classifier gets its name from the softmax function, which squashes the raw class scores into normalized positive values that sum to one so that the cross-entropy loss can be applied [23].

$$L_i = -f_{y_i} + log \sum_j e^{f_j}$$

The next step is to classify the thirty-six letters using the autoencoder training's features. The output of the autoencoder's hidden layer is sent into a softmax classifier, which divides the data into ten groups. The L-BFGS optimization function is used to train the softmax classifier once more. After around 40 iterations, this algorithm converges. We used the remaining 30 images per letter (for 1005 images) as our test set to assess the system's accuracy. The overall system can be seen below:

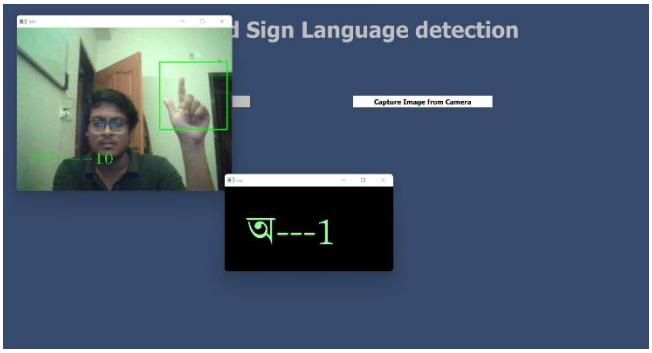


Fig. 9. Overall Preview of our proposed system

V. Experimental Results and Discussion

We went over the details of our implementation of the hand segmentation, unsupervised feature learning, and classification sub-blocks in the preceding sections. This section presents our system's performance in tables and graphs. The accuracy of our classifications is our primary evaluation metric. We want to evaluate how accurate our model is at classifying and predicting the signed letter in an unsegmented image of a person signing a letter. It is hoped to achieve a classification accuracy of roughly 98 percent, similar to many sign language identification scores. We used a learning curve to represent the training error and test error as a function of the Size of the training set as a preliminary diagnosis. The learning curve we acquired is depicted in the graph below, where the X-axis represents the error, and Y-axis represents the Size of the data set:

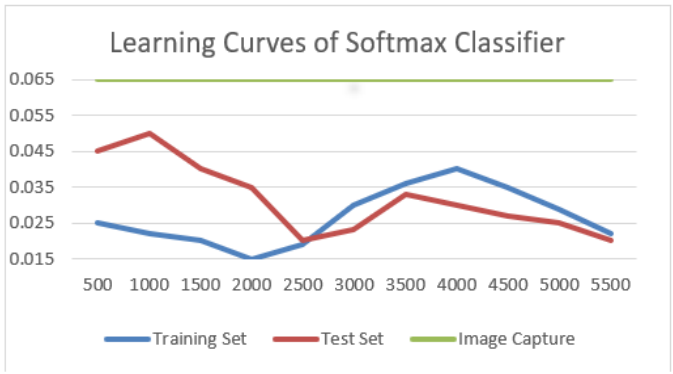


Fig. 10. The learning curve of the sign language recognition system

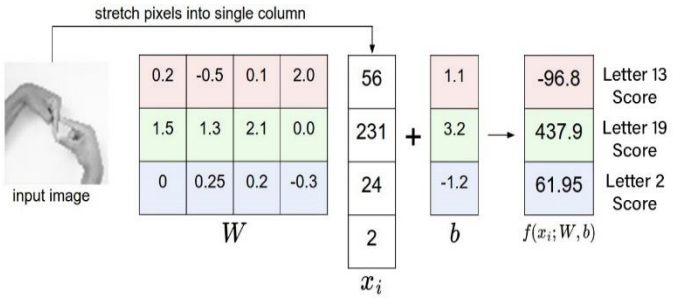


Fig. 11. Analysis of Image Processing (Softmax Classifier)

Conclusion

We used computer vision and machine learning skills to develop a Bangla sign language gesture detection system in this project. We discovered that sometimes simple solutions are more effective than complex solutions. Despite using a sophisticated segmentation technique, the most effective skin masks were extracted using a simple skin segmentation model. We also recognized the time restrictions and challenges of building a dataset from the ground up. Looking back, having a dataset to work with would have been preferable. Some letters were more difficult to classify in our live demonstration, such as "ৎ." Even though our classification approach works very well, as evidenced by the tables and photographs, there is still much room for future research.

Extending the gesture recognition system to all BDSL alphabets and additional non-alphabet gestures would be possible extensions to this project. We believe that by coding in C, we can boost the speed of our system, having chosen MATLAB as the implementation platform. This project's foundation can be used for additional applications, such as commanding robot navigation with hand gestures.

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